

ZPDPatch: Execution-Guided Program Repair from Student Submission Trajectories

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Automated program repair for programming assignments typically treats a student’s latest submission as an isolated buggy program. This snapshot omits the revisions, repeated failures, and partial improvements that led to the current code. We present ZPDPATCH, a repair framework that converts authentic submission trajectories into an execution-ordered portfolio of three repair policies. Verdict and testcase improvements define finite execution-evidence orders; validation execution then selects three checkpoints by exact submodular coverage maximization, while the online controller guarantees observed-test non-regression and optional AST edit-budget compliance. The PROGRESS adapter learns testcase-level Pareto improvements, STRICT learns coarse-verdict improvements, and ANSWER maps failed submissions to the terminal accepted program. We evaluate Qwen2.5-Coder-7B-based adapters on 997 held-out trajectories from 328 test problems drawn from a 492-problem Seen pool, plus 250 trajectories from 54 problem-held-out problems. On Seen problems, the selected portfolio improves repair rate from 30.7% for trajectory-prompted zero-shot to 61.5%; a same-target three-seed Answer control reaches 61.7%, identifying independent checkpoint diversity rather than target heterogeneity as the supported mechanism. On Unseen problems, selection reaches 74.8% RR versus 62.0% for zero-shot and 73.2% for Answer-3Seed. The one-per-relation constraint reduces Seen coverage, whereas budget-indexed unconstrained selection improves the mean Unseen edit-budget frontier by 2.60 points [0.38, 5.03] over Answer-3Seed. A target-matched ablation finds no stable benefit from serializing earlier submissions, and all primary differences use problem-clustered inference.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; • **Applied computing** → **Education**; • **Computing methodologies** → **Natural language generation**.

Additional Key Words and Phrases: automated program repair, programming education, large language models, submission trajectories, execution-guided repair

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1 Introduction

Programming students iteratively write, submit, execute, and revise code. Online judges make this loop scalable, but their feedback is usually limited to coarse verdicts such as Wrong Answer, Runtime Error, or Time Limit Exceeded. A verdict reports failure without explaining whether the latest revision made progress, repeated an earlier defect, or moved away from a viable solution. Instructors cannot inspect every trajectory and author an individualized repair at the scale of modern programming courses.

Automated program repair (APR) offers executable, code-specific feedback. The field spans search-based generate-and-validate repair, learned transformations, and test-guided validation [Gazzola et al. 2019; Le Goues et al. 2012; Monperrus 2018; Weimer et al. 2009]. Systems for programming assignments cluster peer solutions, align a student’s code to a reference, or synthesize a fixed program [Gulwani et al. 2018; Hu et al. 2020; Wang et al. 2018]. More recent systems use language models, execution diagnostics, retrieved examples, and iterative conversations [Joshi et al. 2023; Liu et al. 2024; Orvalho et al. 2025; Yang et al. 2024; Zhang et al. 2024]. Most of these approaches condition on the latest buggy program, possibly augmented with tests or peer programs. The earlier submissions produced by the same student are rarely treated as first-class repair context.

The motivation for using history is straightforward but unverified. Two students may arrive at similar current programs through different revisions. Their histories expose repeated errors,

attempted edits, and states that they demonstrably reached. In educational terms, such reachable improvements resemble the Zone of Proximal Development (ZPD): states attainable with appropriate support but not yet reached independently [Chounta et al. 2017; Cole et al. 1978; Murray and Arroyo 2002]. We therefore ask whether repair can exploit a student’s observed trajectory to produce a useful next program rather than immediately replacing it with an arbitrary correct solution. We use ZPD as a design motivation, not as a measured psychological construct or a claim about learning outcomes.

We introduce ZPDPATCH, illustrated in Figure 1. It automatically converts authentic submission trajectories into three supervised repair datasets. PROGRESS retains verdict improvements and same-verdict transitions whose per-test outcomes improve monotonically. STRICT retains only transitions that improve the online-judge verdict. ANSWER pairs every failed submission with the trajectory’s final accepted program. Three seeds per target relation produce nine independently trained QLoRA checkpoints over the same code language model and prompt. Validation execution selects three complementary checkpoints, which are invoked from narrower to broader supervision. ZPDPATCH executes each candidate, stops on an eligible acceptance, and otherwise selects a non-regressive candidate using test pass rate and AST tree edit distance (TED).

Our evaluation separates three mechanisms that are easy to conflate: learned relational targets, heterogeneous policy composition, and the serialization of prior submissions. The complete system substantially improves over zero-shot on both seen and problem-held-out data, and sequential escalation outperforms each adapter alone. Target-matched retraining finds no stable causal benefit from serializing earlier submissions; both conditions still use targets mined from trajectories, so this null result neither establishes nor refutes relational supervision as the source of the complete-system gain. We therefore test policy heterogeneity directly against three independently trained Answer adapters that share examples, targets, and decoding but differ in training seed.

This paper makes the following contributions:

- We formulate trajectory-mined repair as finite execution-evidence orders plus terminal closure, and audit every materialized supervision transition against those definitions.
- We pose size-three checkpoint choice as exact submodular validation-coverage maximization, derive a finite-family problem-level selection guarantee, and design an execution controller with observed-test non-regression and structural edit-budget compliance by construction.
- We conduct a problem-familiarity-aware evaluation against zero-shot and retrieval-based repair, including the missing same-target three-seed control, patch-budget curves, target-matched history ablations, ordering, and problem-clustered analyses.

2 Background and Related Work

2.1 Structure- and Reference-Based Educational Repair

Educational APR can exploit many programs written for one specification. CLARA clusters and aligns programs before extracting a repair; SARFGEN searches and aligns reference programs; Refactory normalizes refactorings before comparison [Gulwani et al. 2018; Hu et al. 2020; Wang et al. 2018]. Embeddings, solution-space models, and visualization propagate annotations or expose recurring strategies, while multi-function and diversified methods broaden the target beyond one correction [Choi et al. 2021; Choi and Lee 2025; Glassman et al. 2015; Heo et al. 2023; Piech et al. 2015b; Rivers and Koedinger 2017; Yi et al. 2017]. These methods provide strong assignment-specific evidence and establish that execution correctness and relationship to the student’s program are separate repair properties.

LSGen is our reference-rich baseline. It filters historical repair pairs, retrieves by edit information, generates, and may retrieve again from the new state [Dai et al. 2026]. We retain its access to other users' Seen-Train solutions for the target problem. That information is unavailable on genuinely new assignments, so LSGen is not evaluated on problem-held-out Unseen tasks. This separates repository-mature repair from repair using only a learned model, statement, and tests.

2.2 Neural and Large-Language-Model Repair

Neural repair learned transformations from repository histories and context-aware translation before pretrained code models enabled zero-shot and fine-tuned repair [Jiang et al. 2023; Lutellier et al. 2020; Tufano et al. 2019; Xia and Zhang 2022; Yasunaga and Liang 2021]. RING, PyDex, and FastFixer cast educational repair as conditional generation [Joshi et al. 2023; Liu et al. 2024; Zhang et al. 2022, 2024]; retrieval and localization further constrain generation with peer programs or high-precision syntax feedback [Phung et al. 2023; Zhao et al. 2024]. These systems cover programs that do not align cleanly with a reference, but make correctness and patch scope external validation concerns.

Execution-aware methods address those concerns with signals independent of model confidence. SelfAPR learns from test diagnostics; FixEval makes execution the repair-evaluation criterion; CREF carries diagnostics through a repair conversation; counterexample-guided repair combines generation, fault localization, and MaxSAT [Haque et al. 2023; Orvalho et al. 2025; Yang et al. 2024; Ye et al. 2022]. ZPDPATCH likewise executes candidates, but introduces a different unit of supervision and control. Authentic trajectories define nested improvement relations rather than synthetic corruptions or manually assigned repair roles. Independently trained members share a base model, generated candidates remain conditionally independent given authentic observations, and an explicit selector includes the unchanged program. Thus a failed generation cannot force a regression or contaminate another policy's input distribution.

2.3 Adaptive Feedback, ZPD, and Submission Histories

Programming feedback ranges from correctness reports to next-step hints and complete repairs; feedback research emphasizes timeliness, task focus, and actionable information rather than correctness alone [Hattie and Timperley 2007; Keuning et al. 2018; Narciss 2008; Shute 2008]. Tutoring research studies help seeking and assistance at a productive granularity; iSnap and LLM hint systems similarly expose next-step or multi-level support [Aleven and Koedinger 2000; Heickal and Lan 2024; Price et al. 2017; Roest et al. 2024; Xiao et al. 2024]. The ZPD motivates support between independent and assisted performance [Chounta et al. 2017; Cole et al. 1978; Murray and Arroyo 2002; Schulze Bernd and Chounta 2024]. We operationalize only observed reachability: a later improving submission proves that this user reached that program state, not that showing the edit causes learning or is optimal scaffolding.

Submission sequences also support knowledge tracing, which estimates latent mastery from interaction histories [Corbett and Anderson 1994; Ghosh et al. 2020; Piech et al. 2015a]. ZPDPATCH instead orders source-code states using execution evidence; it neither infers mastery nor predicts a learner response. This distinction permits fully automatic repair targets while delimiting the contribution to program behavior rather than educational outcomes.

2.4 Positioning of This Work

Table 1 locates the contribution along evidence and composition axes. Unlike reference alignment and LSGen, ZPDPATCH does not require a same-problem solution repository; unlike prior LLM and iterative repair, its targets are mined as ordered transitions from the same user's authentic

submissions. Its technical contribution is the combination of this execution-evidence order, policy-specific parameterization, and certified sequential selection.

3 Problem Formulation

For problem p , let a student's i -th submission be $s_i = (c_i, v_i)$, where c_i is the complete source code and v_i is the online-judge verdict. A submission trajectory $\tau = [s_1, s_2, \dots, s_n]$ is chronologically ordered and terminates at the first accepted submission. Let d_p contain the problem statement and resource constraints, and let $\mathcal{E}_p$ be the available test suite. Re-executing c_i gives a per-test outcome vector $o_i = (o_i(e))_{e \in \mathcal{E}_p}$.

A repair example consists of problem context d_p , an observed sequence h , and a complete target program y . Adapter-specific rules decide which submissions remain in h and which later submission supplies y ; the two need not be adjacent in the original trajectory. Given $x = (d_p, h)$, a model generates one complete program $\hat{c}$.

We call a later state *reachable* when it occurs in the same student's observed trajectory and satisfies an adapter-specific improvement rule. Reachability is an empirical property of the data, not a guarantee that the transition is minimal, comprehensible, or caused by instructional support. Our design hypothesis is that learning from such transitions can produce repairs that preserve more of the current solution than direct answer generation.

4 Approach

4.1 Design Requirements

Four requirements shape the method: *automatic supervision* recoverable from submissions and tests; *target diversity* spanning intermediate improvement and accepted solutions; *executable validation* independent of model confidence; and *non-regression* when no candidate improves observed behavior. We separate these offline and online decisions. Label relations determine what each adapter learns; execution and selection determine what the system may return. A noisy target can influence generation without automatically overriding a better current program.

4.2 Trajectory Construction

We group submissions by problem and student, sort by time, and stop at first acceptance. After indentation/comment/blank-line normalization, we remove only consecutive duplicates; non-consecutive revisits remain evidence of distinct attempts. Complete source, verdict, and order are preserved, with no maximum history or transition count. Every state is re-executed. Cache keys bind problem, normalized code, test-suite identity, timeout, and memory limit, preventing incompatible reuse. Runner failures receive an explicit most-severe outcome, and Progress construction requires a complete cache.

4.3 Adapter-Specific Supervision

We order verdict severity as $\text{sev}(\text{AC}) = 0$, $\text{sev}(\text{WA}) = 1$, $\text{sev}(\text{TLE}/\text{MLE}) = 2$, $\text{sev}(\text{RE}) = 3$, $\text{sev}(\text{CE}) = 4$, and 5 for internal failures. For equal, non-empty test keys, $o_t \preceq_{\text{sev}} o_r$ iff no test outcome in o_t is more severe than its counterpart in o_r . Test-case progress is the Pareto condition

$$\text{TCProg}(r, t) = \mathbf{1}[o_t \preceq_{\text{sev}} o_r \wedge o_t \neq o_r]. \quad (1)$$

Answer. If s_n is accepted, every preceding non-accepted submission is independently paired with c_n . For $\tau = [S_1, \dots, S_9]$, the rule produces $([S_1], c_9), \dots, ([S_8], c_9)$. Thus, ANSWER learns a direct failed-program-to-answer mapping and receives no multi-submission history.

Table 1. High-level positioning of educational repair approaches. “Same-problem refs.” denotes a database of solutions or repair pairs for the target assignment; “trajectory supervision” denotes training targets mined from ordered submissions by the same user.

Approach family	Refs.	Exec.	Traj.	Stages
Structural alignment [Gulwani et al. 2018; Hu et al. 2020; Wang et al. 2018]	Yes	Opt.	No	No
LLM generation [Joshi et al. 2023; Liu et al. 2024; Zhang et al. 2024]	Opt.	Opt.	No	No
Iterative repair [Orvalho et al. 2025; Yang et al. 2024]	Opt.	Yes	No	Yes
LSGen [Dai et al. 2026]	Yes	Yes	No	Yes
ZPDPATCH	No	Yes	Yes	Yes

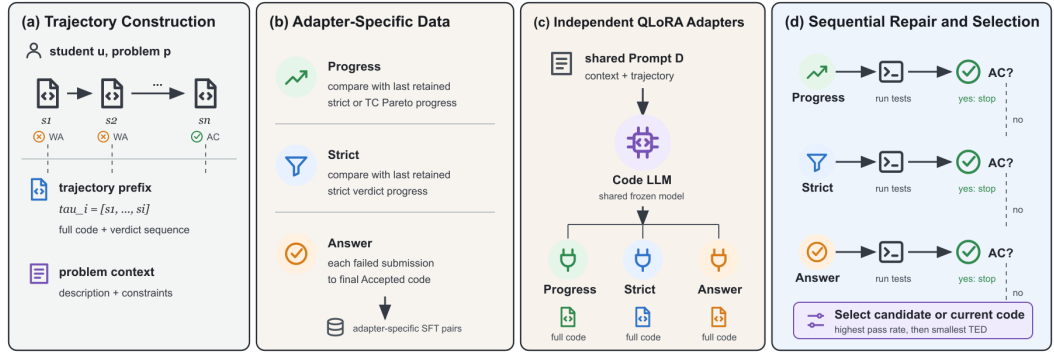


Fig. 1. Overview of trajectory supervision and execution control. The displayed P–S–A route is the one-per-relation control; the deployed system trains three seeds per relation and uses validation execution to select any three checkpoints, so a relation may repeat. Execution outcomes determine early stopping and final selection.

Strict. Starting with $R = [s_1]$, we scan the trajectory and append s_t only when its verdict is strictly better than the last retained submission $s_r = R[-1]$: $\text{sev}(v_t) < \text{sev}(v_r)$. If $R = [r_1, \dots, r_m]$, we create $([r_1, \dots, r_{k-1}], c_{r_k})$ for every $k = 2, \dots, m$. The comparison is against the last retained state, not necessarily the immediately preceding original submission.

Progress. PROGRESS uses the same retained scan but appends s_t when either its verdict strictly improves or $v_t = v_r$ and $\text{TCProg}(r, t) = 1$. It therefore captures monotonic per-test progress that is hidden by an unchanged coarse verdict. Prefix-target construction is otherwise identical to STRICT.

The datasets are constructed independently and may overlap. A strict transition is also a progress transition, and final accepted transitions can occur in both. The objectives nevertheless differ: ANSWER learns direct completion from one failed program, whereas STRICT and PROGRESS learn transitions from retained prefixes.

The common retained-prefix procedure scans $s_2, \dots, s_n$, compares each state with $R[-1]$, and, whenever the corresponding predicate above holds, emits (R, c_t) before appending s_t to R . Comparing with the last retained rather than immediately previous raw state makes R monotonic under the selected relation. A trajectory with m retained states yields all $m - 1$ prefixes, so early transitions are not discarded.

4.4 Execution-Evidence Order

The three rules are not unrelated task labels. They form two nested execution-evidence orders plus a terminal closure. Let the evidence of submission (s) be $(e(s)=v(s),o(s))$ over the fixed testcase universe $\mathcal{E}_p$. Define the strict relation

$$s \prec_S t \iff \text{sev}(v(t)) < \text{sev}(v(s)), \quad (2)$$

and the progress relation

$$s \prec_P t \iff s \prec_S t \vee (v(s) = v(t) \wedge o(t) \prec_\times o(s)), \quad (3)$$

where $o(t) \prec_\times o(s)$ is the strict product order: every testcase outcome is no worse and at least one is better under severity. Strict and Progress retain a chain under $\prec_S$ and $\prec_P$, respectively. Answer is the terminal closure $A(s) = s_n$ from each failed state to the first accepted state; it is intentionally not an intermediate order.

This view makes the target granularity explicit. Strict observes only a projection of execution evidence onto the coarse verdict. Progress refines that projection with the testcase product order. Answer discards intermediate ordering and connects a failed state to the absorbing accepted state. Adapter specialization therefore corresponds to distinct relations over the same trajectory rather than to role words inserted in prompts.

PROPOSITION 1 (FINITE MONOTONE SUPERVISION). *For a problem with T tests, every transition retained by Progress strictly decreases the lexicographic rank $\rho(s) = (\text{sev}(v(s)), \sum_{e \in \mathcal{E}_p} \text{sev}(o(s, e)))$. Consequently, retained Progress chains contain no cycles and have length at most $6(5T + 1)$. Strict chains satisfy the same result through their first rank component.*

PROOF. A Strict transition decreases the first component, independently of the second. A same-verdict Progress transition is Pareto non-worsening and strict on at least one testcase, so it preserves the first component and decreases the sum. Thus every retained transition strictly decreases ρ . There are six verdict-severity values and $5T + 1$ possible sums, which gives the finite bound. $\square$

Because $\prec_S \subseteq \prec_P$, every Strict target event is also a Progress event. Moreover, for any failed state, the first Progress-improving state cannot occur after the first Strict-improving state, and acceptance closes both relations. These inclusion facts determine the fine-to-broad policy order; they are construction invariants rather than claims about learned behavior.

Order versus scalar reward. A passed-test count can improve while breaking a previously passing testcase; Equation 1 refuses this compensation. Coarse verdict is the first coordinate and non-compensatory testcase severity refines equal verdicts, so the rank proves termination while the product order records which observations justify a transition. Execution certifies retained supervision, not model behavior on new inputs; the online controller must still test every candidate and retain the original fallback. The order is defined only over one fixed testcase universe and severity map, so cache keys bind tests and resource limits and Progress fails closed on incomplete vectors. Otherwise a missing test could turn a regression into apparent Pareto progress.

4.5 Prompt and Adapter Training

All adapters share one role-neutral prompt: statement and limits, chronologically retained complete submissions with verdict/outcome/edit summaries, and the current submission requesting one complete solution in the dataset's language and format. No instruction names a policy or asks for a particular edit size, so specialization can arise only from supervision. Only assistant target tokens contribute to loss.

For adapter $a \in \{\text{prog, strict, ans}\}$, let $\mathcal{D}_a$ be its dataset, x_i the prompt, and y_i the target tokens. We minimize the target-only supervised fine-tuning objective

$$\mathcal{L}_a = -\frac{1}{|\mathcal{D}_a|} \sum_{(x_i, y_i) \in \mathcal{D}_a} \frac{1}{|y_i|} \sum_{k=1}^{|y_i|} \log P_{\theta, \phi_a}(y_{i,k} \mid y_{i, < k}, x_i), \quad (4)$$

where θ is the frozen base model and ϕ_a is an adapter's QLoRA parameter set. Prompt tokens are masked from the loss. The adapters are trained and stored independently.

4.6 Validation-Selected Execution Portfolio

Training loss ranks checkpoints within one target distribution, but it does not measure the cross-policy repair complementarity used at deployment. We therefore select the portfolio using execution on validation problems. Let $V = V_P \dot{\cup} V_S \dot{\cup} V_A$ be independently trained Progress, Strict, and Answer checkpoints and let $R_v \subseteq U$ be the validation instances repaired by checkpoint v . For a portfolio S , define execution coverage

$$f(S) = \left| \bigcup_{v \in S} R_v \right| / |U|. \quad (5)$$

The deployed search family is $\mathcal{F}_3 = \{S \subseteq V : |S| = 3\}$. It allows validation evidence to choose both target relations and training seeds rather than assuming that one checkpoint per relation is optimal. We retain $\mathcal{F}_{\text{rel}} = \{S \in \mathcal{F}_3 : |S \cap V_a| = 1 \ \forall a\}$ as the direct relation-heterogeneity control. We sample one trajectory per validation problem by a fixed hash before any execution, preventing high-volume assignments from dominating selection. In addition to unrestricted f , let R_v^B contain repairs whose AST TED is at most B , and predeclare $\mathcal{B} = \{5, 10, 20, 40, 80, 160\}$. Define $f_B(S) = |\bigcup_{v \in S} R_v^B| / |U|$. The budget-aware objective $f_B(S) = |\mathcal{B}|^{-1} \sum_{B \in \mathcal{B}} f_B(S)$ averages the complete coverage curve for deployments that require one budget-agnostic triple. When the budget is declared before a query, the aligned controller instead freezes the validation maximizer for that budget and invokes its triple under the corresponding gate. We exactly enumerate all $\binom{9}{3} = 84$ members of $\mathcal{F}_3$ and the 27 relation-constrained triples for unrestricted, budget-agnostic, and each budget-indexed objective. The three Answer checkpoints form the prespecified same-target control. Test outcomes never enter selection.

PROPOSITION 2 (COVERAGE STRUCTURE AND EXACT SELECTION). *The objectives f , every f_B , and $f_{\mathcal{B}}$ are normalized, monotone, and submodular. Exhaustive enumeration returns their global optima over $\mathcal{F}_3$, and the parallel enumeration of $V_P \times V_S \times V_A$ returns the global optima under the relation constraint.*

PROOF. Normalization and monotonicity follow from set union. For $S \subseteq T$ and checkpoint $v \notin T$, the marginal gain is $|R_v \setminus \bigcup_{u \in S} R_u| / |U|$, which cannot be smaller than the same quantity with T replacing S ; hence f is submodular. Enumeration evaluates every feasible triple. Each budgeted coverage term has the same set union form, and a nonnegative sum of submodular functions is submodular. The two finite enumerations inspect every member of their respective feasible sets. $\square$

Exact optimization does not by itself guarantee that the validation winner is the best population portfolio. The finite search space nevertheless permits a problem-level guarantee. Let $Z_1, \dots, Z_m$ be independent validation problems drawn from the deployment distribution, let $h_S(Z) \in \{0, 1\}$ indicate whether portfolio S repairs the sampled state for problem Z , and let $F(S) = \mathbb{E}[h_S(Z)]$.

THEOREM 1 (FINITE-PORTFOLIO SELECTION GUARANTEE). *For a frozen family $\mathcal{F}$ of M candidate portfolios, let $\hat{S}$ maximize empirical problem-level repair coverage on m validation problems and let S^**

maximize F . With probability at least $1 - \delta$,

$$F(S^*) - F(\widehat{S}) \leq \sqrt{\frac{2 \log(2M/\delta)}{m}}. \quad (6)$$

The same statement holds for a fixed edit budget after replacing h_S by the budget-gated repair indicator; simultaneous selection over q declared budgets replaces M by Mq .

PROOF. For any fixed S , Hoeffding's inequality bounds the deviation between its empirical and population coverage by $2e^{-2m\epsilon^2}$. A union bound over the M portfolios makes all deviations at most $\epsilon = \sqrt{\log(2M/\delta)/(2m)}$ simultaneously. On that event, $F(S^*) \leq \widehat{F}(S^*) + \epsilon \leq \widehat{F}(\widehat{S}) + \epsilon \leq F(\widehat{S}) + 2\epsilon$, which gives Equation 6. The budgeted cases use the same binary indicator and a union bound over the declared objectives. $\square$

The theorem makes the unit of selection explicit: sampling one state per validation problem supports a problem-distribution claim, whereas treating all trajectories as independent would not. Its i.i.d. premise is an idealization; we therefore freeze the selected portfolios and report independent test results with problem-clustered intervals rather than using the bound as an effect estimate.

4.7 Execution-Guided Sequential Repair

At inference, we invoke the three validation-selected checkpoints in fine-to-broad relation order, breaking same-relation ties by their frozen member order. Each checkpoint receives the same observed student trajectory and produces one complete program; generated candidates are never inserted into another checkpoint's prompt. This independence makes the outputs a policy portfolio rather than a synthetic trajectory whose later states may lie outside the training distribution. In unrestricted mode, an accepted candidate stops the sequence; otherwise executed candidates are compared by the certified selector below. The budgeted mode additionally continues past accepted candidates that exceed the declared structural limit. The order is fixed before evaluation and tested independently in RQ3–RQ6. We also evaluate the previously plausible alternative of appending generated execution feedback as a controlled ablation.

If all candidates fail, the selector includes the current program c_i as a fallback:

$$C = \{c_i, c^{(1)}, c^{(2)}, c^{(3)}\}. \quad (7)$$

Let $\text{Pass}(c)$ be the fraction of tests passed and $\text{ted}(c_i, c)$ the Python AST tree edit distance. Selection is lexicographic:

$$c^* = \arg \max_{c \in C} (\text{Pass}(c), -\text{ted}(c_i, c)). \quad (8)$$

The selector first maximizes executed correctness and then minimizes change. If no candidate improves on c_i , the unchanged fallback has TED zero and the system abstains. Sequentially generating, executing, and stopping at the first accepted candidate preserves conditional independence given the authentic observed trajectory; if all fail, every executed candidate remains available to Equation 8.

Certified selection and cost. Including c_i in Equation 8 makes observed-test non-regression an invariant: an early return has pass rate one, and otherwise the maximum cannot score below its fallback. With a structural budget B , the gate G_B discards undefined or over-budget TED, continues past an over-budget acceptance, and selects from the eligible candidates plus c_i . Thus every returned change has pass rate at least $\text{Pass}(c_i)$ and $\text{ted}(c_i, c^*) \leq B$; deployed coverage is exactly the validation set-union objective f_B . These guarantees concern observed tests and edit compliance, not hidden-test correctness or learned relational behavior.

With cached outcomes, constructing a length- n trajectory over T tests costs $O(nT)$ time and $O(n)$ retained-state storage; Answer construction is $O(n)$. Inference performs one to three generations and executions. Early acceptance can reduce this cost, while the cache avoids repeated execution of identical candidates without changing the worst case.

5 Experimental Design

5.1 Research Questions

RQ1: Repair Effectiveness. How effectively does ZPDPATCH repair held-out trajectories from problems observed during training, compared with zero-shot and retrieval-based repair?

RQ2: Trajectory Conditioning. Does access to the full submission prefix improve repair over training and inference with only the current code?

RQ3: Adapter Specialization. Do PROGRESS, STRICT, and ANSWER exhibit different repair coverage and patch size?

RQ4: Sequential Escalation. Does execution-guided composition improve effectiveness without always invoking all adapters?

RQ5: Problem Generalization. Does ZPDPATCH retain an advantage on problems completely excluded from training?

RQ6: Robustness and Transfer. Are the conclusions robust to training seed, user overlap, problem-level dependence, policy ordering, and a student-held-out Java course dataset?

5.2 Dataset Construction and Splits

We combine Python submissions, statements, metadata, and test cases for problems in Project CodeNet’s Python800 benchmark [Puri et al. 2021]. We retain trajectories with at least three submissions, at least one non-accepted state before the first acceptance, and a final accepted program present in the Python800 reference set. We remove post-acceptance submissions and consecutive normalized duplicates. Users must appear in at least three problems and each retained problem must contain at least two trajectories. The refined corpus contains 546 problems, 21,454 trajectories, 99,432 submissions, and 10,088 tests.

Before splitting, we tokenize every possible ANSWER, STRICT, conservative PROGRESS, and final-evaluation prompt-target configuration. If any configuration exceeds 4,096 tokens, we exclude the entire trajectory rather than truncating code or history. This removes 2,896 trajectories and leaves 18,558. No later max_history, transition-count, code, or test truncation is applied.

We sort problems by eligible trajectory count and assign the top 90% (492 problems) to *Seen* and the remaining 54 low-resource problems to *Unseen*. Within every *Seen* problem, we reserve at least one trajectory for Train, Validation, and Test, then obtain a deterministic 80:10:10 trajectory split. All 260 *Unseen* trajectories remain problem-held-out test data. Table 2 reports the resulting corpus. Users may occur in more than one split; RQ5 is problem-held-out, not user-held-out.

Adapter construction yields the training and validation sizes in Table 3. Final evaluation uses the prefix ending at the last non-accepted submission and the final accepted program as the oracle. We retain only cases for which re-execution confirms the current program as non-accepted and the oracle as accepted, leaving 997 *Seen* instances from 328 of the 492 *Seen* problems and 250 *Unseen* instances from all 54 held-out problems. The remaining *Seen* problems still contribute Train or Validation trajectories but have no Test trajectory satisfying both execution checks; this is why the abstract’s test-problem count is smaller than the *Seen* pool.

To test the educational and language scope directly, we add TIKTOC’s testcase-augmented CodeWorkout release, which contains authentic CS1 Java submissions and per-test outcomes [Duan

Table 2. Dataset statistics after whole-trajectory context filtering. “Prefix” counts possible raw prefixes before adapter-specific filtering.

Group	Role	Problems	Tests	Traj.	Sub.	Prefix	Users
Seen	Train			14,638	58,872	44,234	4,072
	Validation	492	9,446	1,830	7,283	5,453	1,283
	Test			1,830	7,068	5,238	1,291
Unseen	Test	54	642	260	965	705	237

Table 3. Adapter supervision datasets. Validation examples are excluded from training.

Adapter	Train examples	Validation examples
PROGRESS	21,416	2,685
STRICT	16,973	2,116
ANSWER	40,454	4,965

et al. 2025]. We apply the same first-acceptance, duplicate, minimum-length, and 4,096-token whole-trajectory rules, then split by student rather than trajectory. The resulting external corpus has 1,125 trajectories from 224 students over all 17 exercises, with zero student overlap: 786/158/181 train/validation/test trajectories from 157/34/33 students. Every split contains all 17 exercises. External training contains 3,470 Answer, 1,110 Strict, and 1,386 Progress examples; validation contains 730/230/276. We train the same model and hyperparameters from the same base checkpoint. For each relation, seeds 2027/2028/2029 create nine independent checkpoints. We execute all nine on the 158 student-disjoint validation states, exactly enumerate all 84 size-three and 27 one-per-relation portfolios under Equation 5, and freeze both winners before testing. The 181 student-held-out final states compare both with the mandatory same-target Answer-3Seed control; no CodeWorkout test outcome enters checkpoint or portfolio selection.

5.3 Baselines

Zero-shot. The zero-shot baseline uses the same Qwen base model and prompt as ZPDPATCH without fine-tuning. If a candidate fails, its verdict and number of passed tests are added to the conversation before the next attempt. It receives at most three generations, matching ZPDPATCH’s maximum budget. Because it receives the full trajectory, comparison with zero-shot measures the complete effect of learned adapters and sequential composition, not history in isolation.

LSGen. LSGen is an iterative retrieval-based solution generator [Dai et al. 2026]. We preserve its repair-pair filtering, edit-vector retrieval, diff-guided generation, and re-retrieval, while replacing hard-coded infrastructure with our common dataset and runner. For a Seen query, the retrieval database contains buggy/correct pairs from other Seen-Train trajectories of the same problem; the same student and example are excluded. LSGen retrieves five pairs and performs at most three iterations. We do not evaluate it on Unseen problems because exposing same-problem correct programs would violate problem holdout.

All approaches use the same base LLM, public tests, 2.5-second per-test timeout, 2GB memory limit, and a maximum of three candidate generations. Final validation-selected, external-transfer, and budget-controller runs cap every new candidate at 4,096 new tokens; reused control candidates are audited against the same bound.

5.4 Implementation

We use Qwen2.5-Coder-7B-Instruct [Hui et al. 2024] and train each adapter for one epoch with 4-bit QLoRA [Dettmers et al. 2023]. The base weights use NF4 double quantization and BF16 computation. LoRA targets the q, k, v, o attention projections and gate/up/down MLP projections with rank 16, scaling 32, and dropout 0.05. Optimization uses paged 8-bit AdamW, learning rate 2×10^{-4} , cosine scheduling, 0.03 warmup ratio, and gradient checkpointing. Both canonical and CodeWorkout training use micro-batches of 2 with eight accumulation steps, for an effective batch of 16. Seed 2027 is primary; seeds 2028 and 2029 repeat all adapter trainings and evaluation with identical within-dataset data, hyperparameters, and greedy decoding. We select the lowest validation-loss checkpoint and run on one NVIDIA RTX 5090.

OpenAI Codex was used interactively to propose control and validation-selection design alternatives, implement dataset builders, orchestrate experiments, develop statistical-analysis scripts, and assist manuscript drafting. The authors selected the final design and acceptance criteria, executed every job on author-controlled infrastructure, reviewed code changes, and checked reported values against saved machine-readable outputs. AI-produced text was not treated as empirical evidence, and every cited work was checked against a retrievable source.

An isolated runner executes original and generated programs: Python800 uses the Python judge and CodeWorkout compiles Java 21 in a network-disabled container. We cache outcomes by problem, code, test set, timeout, and memory configuration and reuse the cache across supervision construction and evaluation. Within an evaluation split, every policy is bound to the same cached current-program verdict and pass rate; only a previously unseen generated candidate is executed, which prevents timeout-boundary noise from changing the baseline across portfolio members. The Java harness reproduces all 181 recorded accepted test programs before model evaluation.

5.5 Ablations

For RQ2, we construct adapter-specific STRICT and PROGRESS examples for both Seen and Unseen tests. *Full Trajectory* uses each retained prefix. *Current Code Only* retrains an adapter on identical examples and targets after removing every submission before the current one; its displayed position is reset to S_1 to prevent index leakage. *ANSWER* is excluded because its inputs already contain one submission. This design isolates the presence of history while holding targets, counts, model, and hyperparameters fixed.

For RQ3, each adapter independently generates one candidate for the same 997 Seen instances. For RQ4, we first compose the seed-2027 candidates with PROGRESS-STRICT-ANSWER early stopping and fallback selection, then evaluate the final validation-selected size-three portfolio. Four controls separate target heterogeneity from repeated calls, prompt-state effects, training randomness, and the relation constraint. *Answer×3* invokes the same Answer adapter up to three times, appending generated execution feedback so that deterministic greedy decoding can produce a non-identical later candidate; repeating the identical prompt would collapse exactly to Answer once. *Generated Feedback* applies the same update protocol to the heterogeneous P-S-A policies. *Answer-3Seed* independently trains three Answer adapters on the identical 40,454 training and 4,965 validation examples with identical targets and hyperparameters, changing only seeds 2027/2028/2029; all three receive the same authentic evaluation trajectory without generated feedback. *Validation Relation-Constrained* selects one checkpoint per relation, while the primary *Validation Execution Portfolio* selects any three of all nine checkpoints on the same problem-balanced validation execution; both are frozen before test. Every independent policy remains conditioned on the same authentic trajectory. All configurations share the maximum three-generation budget, execution limits, early stopping, and fallback selector. We audit matching Answer training metadata and pairwise-distinct

adapter-weight hashes. Effectiveness-only controls may omit AST TED only when failed pass-rate ties are resolved by earliest candidate, which cannot change PR, RR, or IR; Answer-3Seed and the validation-selected portfolios instead use the complete production selector so patch size can also be compared.

5.6 Metrics and Statistical Analysis

For M instances, let c_m be the current program and $\hat{c}_m$ the selected result. $\text{Pass}(c) \in [0, 1]$ is the test pass fraction. Pass Rate (PR) averages $\text{Pass}(\hat{c}_m)$; Repair Rate (RR) averages $1[\text{Pass}(\hat{c}_m) = 1]$; Improvement Rate (IR) averages $1[\text{Pass}(\hat{c}_m) > \text{Pass}(c_m)]$.

For successfully repaired, parseable programs, $\text{TED}_{B \rightarrow F}$ measures AST distance from current to fixed code and $\text{TED}_{F \rightarrow O}$ from fixed code to the recorded accepted oracle, using the standard unit-cost ordered-tree edit formulation [Zhang and Shasha 1989]. Legacy fixed-policy reports use ZSS; validation-selected portfolios and exhaustive budget replay use the equivalent APTED implementation [Pawlik and Augsten 2016], cross-checked against ZSS on identical labeled ASTs. Because node-count difference lower-bounds unit-cost TED, pairs whose difference already exceeds the largest declared budget are stored only as > 160 , which preserves every budget decision. RQ2 uses $\text{TED}_{F \rightarrow T}$, where T is the identical adapter-specific recorded target shared by Full and Current configurations.

We report Wilson 95% intervals for RR and IR [Wilson 1927]. Paired RR comparisons use two-sided exact McNemar tests [McNemar 1947], with Holm correction within each planned comparison family [Holm 1979]. Because trajectories from one problem are dependent, our primary robustness analysis resamples problems as clusters 10,000 times with seed 2027 and reports intervals for paired RR, PR, and IR differences. We additionally report a problem-balanced estimand that gives every problem equal weight, with a problem bootstrap interval. Paired TED differences use 10,000 instance bootstrap resamples on jointly repaired inputs; both procedures follow the nonparametric bootstrap principle [Efron 1979]. Unless stated otherwise, percentages are absolute percentage points.

6 Results

6.1 RQ1: Repair Effectiveness

Table 5 summarizes RQ1 and RQ5. On Seen problems, the validation-selected execution portfolio repairs 613 of 997 programs (61.5%), compared with 306 (30.7%) for Zero-shot. Their Wilson RR intervals are [58.4, 64.5] and [27.9, 33.6], respectively. ZPDPATCH alone repairs 342 instances, whereas Zero-shot alone repairs 35 (exact McNemar $p < 10^{-63}$). More importantly for dependent trajectories, the 30.8-point paired RR gain remains under problem-cluster bootstrap [26.4, 35.1]; the equal-problem-weight gain is 26.1 points [21.2, 30.9]. PR increases by 11.07 points [9.16, 13.02] and IR by 25.2 points [20.8, 29.4] under the same clustered analysis.

The legacy LSGen controller achieves the highest Seen execution correctness: 88.0% PR, 80.2% RR, and 82.4% IR. It alone repairs 267 instances, while ZPDPATCH alone repairs 80 (exact McNemar $p < 10^{-23}$); the clustered RR difference is -18.8 points $[-23.0, -14.5]$. The tradeoff is patch size. ZPDPATCH's mean $\text{TED}_{B \rightarrow F}$ is 21.05, versus 27.25 for Zero-shot and 88.27 for LSGen. Among jointly repaired parseable instances, ZPDPATCH reduces TED by 8.46 relative to Zero-shot (95% CI [5.05, 12.55]) and by 43.34 relative to LSGen (95% CI [38.47, 48.44]). RQ6 replaces the legacy early-stop replay with an always-three LSGen controller before making the final budget-frontier comparison.

Table 4 operationalizes this tradeoff rather than treating TED as evidence of pedagogical quality. Under a maximum structural edit budget, ZPDPATCH has substantially higher repair coverage through $B = 80$; LSGen crosses over only at the broadest reported budget. At $B = 20$, for example, the

gate returns an accepted repair for 39.9% of inputs with ZPDPATCH versus 10.0% with LSGen, a +29.9-point problem-clustered difference [26.2, 33.7]. The comparison therefore supports a constrained deployment claim: retrieval has higher unrestricted coverage, whereas the trajectory-trained portfolio covers more inputs when large structural replacement is disallowed.

Paired outcomes show complementary success rather than marginal exchanges. On Seen, ZPDPATCH and Zero-shot repair 271 jointly, with 342 versus 35 unique repairs; against LSGen the corresponding counts are 533 jointly and 80 versus 267 uniquely. On Unseen, ZPDPATCH and Zero-shot repair 142 jointly, with 45 versus 13 unique repairs. Thus LSGen has higher coverage, but neither Seen success set contains the other.

PR and IR confirm that the RR gain is accompanied by partial behavioral improvement. Patch distance separates repair from replacement: mean current-to-fix TED is 22.8% below Zero-shot and 76.2% below legacy LSGen on each method's successful set, while the jointly repaired paired analyses above establish the same direction without success-set confounding.

6.2 RQ2: Trajectory Conditioning

Table 6 shows no consistent advantage for Full Trajectory. On Seen STRICT examples, Full raises PR by 0.7 points but lowers RR and IR by 0.7 and 1.3. On Seen PROGRESS, it lowers PR, RR, and IR by 1.0, 0.4, and 0.6. On Unseen data, Full is worse for STRICT but better for PROGRESS. Target TED is also mixed: Full is closer to the recorded target for Seen PROGRESS and Unseen STRICT, but farther in the other conditions. None of the four paired RR differences is significant after Holm correction (minimum adjusted $p = 0.313$).

Each Full–Current pair holds training examples, targets, architecture, hyperparameters, and evaluation inputs fixed; only earlier submissions are removed and position leakage reset. Direction changes across conditions, clustered RR intervals include zero (Unseen Strict reaches zero at its upper endpoint), and target TED is mixed. RQ1 can therefore establish complete-system utility while RQ2 rejects history serialization as a stable causal explanation.

6.3 RQ3: Adapter Specialization

Table 7 exposes a coverage-size tradeoff. ANSWER has the highest PR, RR, and IR, but also the largest mean patch at TED 20.22. PROGRESS has lower coverage but the smallest patch at TED 15.85. STRICT and PROGRESS have statistically indistinguishable RR (adjusted $p = 0.857$). ANSWER repairs 5.2 points more than PROGRESS and 5.5 points more than STRICT; problem-cluster intervals are [2.5, 7.9] and [2.9, 8.0], respectively. The ordered increase in patch size is consistent with broader targets, but STRICT does not form a clearly distinct coverage regime.

Coverage specialization is weak between PROGRESS and STRICT, but patch-scope specialization is ordered: their mean current-to-fix TEDs are 15.85, 17.45, and 20.22 through Answer, while Answer repairs 5.2–5.5 points more. Oracle distance follows the same pattern. These are distributional effects rather than decoding constraints; hard non-regression enters only through selection.

6.4 RQ4: Sequential Escalation

Independent-policy escalation reaches 59.5% RR, 9.4 points above the strongest individual adapter, ANSWER; its problem-cluster interval [7.6, 11.3] excludes zero. Gains over PROGRESS and STRICT are 14.6 [12.3, 17.0] and 14.9 [12.8, 17.1] points, respectively. The increase appears in PR and IR as well, so the portfolio is not trading complete repairs for weaker partial outputs.

Progress accepts 447 inputs, Strict accepts 60 more, and 490 reach Answer. The resulting 2,037 generations average 2.04 per input, avoiding 954 (31.9%) of an always-three budget. Final selection returns the unchanged program 329 times, making certified abstention a substantial behavior rather than a corner case.

Table 4. Seen repair rate (%) after the same AST-TED deployment gate. The last column is ZDPATCH minus LSGen with a 95% problem-cluster interval.

Budget B	ZDPATCH	LSGen	Difference [95% CI]
5	20.2	3.3	+16.9 [14.2, 19.7]
10	29.7	5.8	+23.9 [20.7, 27.2]
20	39.9	10.0	+29.9 [26.2, 33.7]
40	48.6	25.7	+22.9 [19.4, 26.5]
80	53.9	47.3	+6.5 [2.7, 10.4]
160	55.8	65.7	-9.9 [-14.1, -5.7]

Table 5. Main repair results. PR, RR, and IR are percentages; TED is computed on successfully repaired, parseable programs.

Split	Method	PR	RR	IR	$TED_{B \rightarrow F}$	$TED_{F \rightarrow O}$
Seen	Zero-shot	76.3	30.7	43.7	27.25	27.67
	LSGen	88.0	80.2	82.4	88.27	83.23
	ZDPATCH	87.4	61.5	68.9	21.05	21.64
Unseen	Zero-shot	75.3	62.0	68.8	17.46	15.80
	ZDPATCH	84.9	74.8	79.6	11.40	11.96

Table 6. RQ2 trajectory-conditioning results. PR, RR, and IR are percentages. Current and Full use identical examples and targets.

Split	Adapter	Input	N	PR	RR	IR	$TED_{F \rightarrow T}$
Seen	STRICT	Current	2,151	56.8	31.2	54.6	40.23
	STRICT	Full	2,151	57.5	30.5	53.3	41.05
	PROGRESS	Current	2,674	58.4	26.3	49.7	36.23
	PROGRESS	Full	2,674	57.4	25.9	49.1	33.15
Unseen	STRICT	Current	322	66.7	56.2	71.4	20.20
	STRICT	Full	322	65.4	53.4	70.8	19.27
	PROGRESS	Current	378	65.7	52.1	66.4	17.63
	PROGRESS	Full	378	66.5	53.7	68.3	18.16

Table 7. RQ3 individual-adapter results on 997 Seen instances.

Adapter	PR	RR	IR	$TED_{B \rightarrow F}$	$TED_{F \rightarrow O}$
PROGRESS	73.4	44.8	52.0	15.85	15.89
STRICT	74.8	44.5	51.2	17.45	18.38
ANSWER	77.5	50.1	57.5	20.22	20.61

Answer to RQ4. Execution-guided escalation improves both complete repair and partial improvement over every single adapter. Repeating Answer with generated feedback reaches 55.2% RR, so additional calls recover 5.0 points over Answer once but remain 4.31 points below the relation-heterogeneous portfolio. The stricter target-matched control reverses that comparison: three

Table 8. RQ4 sequential and validation-selected portfolios on 997 Seen instances. Selection uses one validation trajectory per problem; test outcomes are not used.

Configuration	PR	RR	IR
Answer×3	84.1	55.2	62.4
Answer-3Seed	87.0	61.7	69.1
Generated Feedback	84.9	56.6	64.6
Fixed P-S-A	86.5	59.5	67.0
Validation relation-constrained	85.8	57.3	65.4
Validation unconstrained	87.4	61.5	68.9

independently trained Answer adapters reach 61.7% RR, exceeding Independent Policies by 2.21 points with a problem-cluster interval [0.11, 4.20] (67 Answer-3Seed-only versus 45 independent-policy-only repairs; McNemar $p = 0.0467$). On jointly repaired programs, Independent Policies makes smaller patches by 1.56 TED [0.36, 2.79], but it does not recover the coverage difference.

Validation execution does not rescue the relation constraint on Seen test. The unrestricted relation-constrained winner obtains 57.3% RR, 4.41 points below Answer-3Seed [-6.54, -2.24]. The unconstrained winner (Answer-2027, Answer-2028, Progress-2027) reaches 61.5%, statistically indistinguishable from Answer-3Seed at -0.20 points [-1.83, 1.46]. This control is decisive: checkpoint complementarity transfers, but forcing one member from every relation reduces coverage. Early stopping still avoids a fixed three-generation cost, and Equation 5 remains the test-independent mechanism for selecting the deployed checkpoints.

Under the six predeclared budgets, choosing a separate validation winner for each declared budget gives the unconstrained family +0.75 mean points over Answer-3Seed [-0.29, 1.86] on Seen. The only individually conclusive budget is TED 20 (+1.81 [0.53, 3.16]); the relation-constrained family averages -0.70 [-2.02, 0.71]. Thus the CodeNet evidence supports budget-indexed validation selection at a specific operating point, not a general coverage benefit from relation heterogeneity.

6.5 RQ5: Problem Generalization

On 250 Unseen instances, ZPDPATCH obtains 84.9% PR, 74.8% RR, and 79.6% IR, versus 75.3%, 62.0%, and 68.8% for Zero-shot. ZPDPATCH alone repairs 45 cases and Zero-shot alone repairs 13 (McNemar $p = 3.01 \times 10^{-5}$). Problem-cluster intervals exclude zero for RR (+12.8 points [6.83, 18.75]), PR (+9.56 [5.66, 13.64]), and IR (+10.8 [4.76, 16.95]); the equal-problem-weight RR gain is 12.38 points [6.01, 18.62]. ZPDPATCH also reduces mean successful-patch TED from 17.46 to 11.40. Among 122 jointly repaired parseable instances, its patch is 6.41 TED units smaller on average (95% CI [3.43, 9.89]).

Against the weaker repeated-Answer-with-feedback control on Unseen data, Independent Policies reaches 73.2% versus 68.4% RR, with 15 portfolio-only and three control-only repairs (McNemar $p = 0.00754$). The independent Answer-3Seed control, however, also reaches 73.2%; its problem-clustered difference from Independent Policies is 0.0 points [-3.2, 3.1] and McNemar $p = 1.0$. The problem-held-out result therefore supports execution-selected checkpoint diversity but does not identify target-relation heterogeneity as its source.

The frozen validation winners reach 74.8% Unseen RR for both unrestricted relation-constrained and unconstrained selection, +1.60 points over Answer-3Seed; both unrestricted intervals include zero. Budget-indexed unconstrained selection is stronger across the declared frontier: its mean coverage difference is +2.60 points [0.38, 5.03], whereas the relation-constrained mean is +1.60 [-1.07, 4.44]. This independently confirms the budget-selection effect but again does not support the one-per-relation constraint.

The 45 favorable versus 13 unfavorable discordant pairs against Zero-shot show complementary successes rather than dominance. Absolute Unseen RR is higher for both methods, but the split was defined by trajectory availability rather than randomized difficulty. The valid estimand is the paired within-Unseen difference, which remains positive without same-problem training.

Answer to RQ5. ZPDPATCH’s advantage over Zero-shot persists on problems completely excluded from adapter training. Because the Unseen group deliberately contains lower-resource problems, its higher absolute RR must not be interpreted as evidence that unseen problems are easier or that familiarity harms repair.

6.6 RQ6: Robustness and External Transfer

Implementation conformance. An independent artifact audit checks definitions rather than model quality and is therefore not treated as an empirical result. Answer’s 40,454 examples all pair one failed source with an accepted target; all 19,981 Strict and 29,920 Progress transitions satisfy their declared relation; and all 16,973 unique Strict target events occur among the 21,416 Progress events. Across 997 Seen and 250 Unseen outputs, selected pass rate is never below the current program. These are implementation checks of the formal construction and selector, not a separate empirical claim about learned relations.

Because users may cross splits, we additionally stratify test instances by whether their user identifier occurs in any adapter-training set. Among 155 Seen instances from 150 training-disjoint users, ZPDPATCH reaches 71.6% RR versus 31.6% for Zero-shot, a paired +40.0-point gain with problem-cluster interval [30.7, 48.8]; the corresponding overlap stratum has 842 instances and a +26.7-point gain [22.6, 31.0]. Thus the Seen effect is not confined to repeated users. On Unseen problems, the 69 user-disjoint instances retain a +11.6-point paired difference, but their cluster interval [−1.5, 25.4] includes zero; this smaller subgroup does not establish user-disjoint problem generalization on its own. Strata are observational, so differences between their absolute rates are not causal estimates of memorization.

Complete retraining with seeds 2027/2028/2029 yields Seen RR 59.5/57.6/59.0% (mean 58.7, SD 1.0 points) and Unseen RR 73.2/74.0/70.4% (mean 72.5, SD 1.9). Mean PR/IR are 86.2/66.2% on Seen and 83.5/78.4% on Unseen. Every Seen run exceeds the earlier Generated Feedback configuration’s 56.6% RR. This cross-run comparison bounds QLoRA variation but does not replace the paired seed-2027 comparison, which alone holds trained adapters fixed while changing candidate dependence.

Finally, replaying all six policy orders on fixed candidates leaves RR at 59.5%. Progress–Strict–Answer has the lowest mean breadth (1.391) and successful-patch TED (21.10); Answer-first saves about 0.12 generations but selects broader policies and patches. Separately generated-feedback, repeated-Answer, and Answer-3Seed controls test prompt-state updates, repeated calls, and independent-training diversity.

Answer to RQ6. Problem-clustered inference preserves the principal effects, three full training runs bound seed variation, and exhaustive order replay identifies the selected order as the least-broad, smallest-patch endpoint among fixed-candidate portfolios. The conformance audit separately confirms that saved supervision and selector outputs satisfy the implemented specifications; it is not evidence of learned behavior.

6.7 Cross-RQ Synthesis

Together, the RQs distinguish properties of the data construction from causes of test-time coverage. The execution-evidence orders are exactly satisfied by the materialized supervision, but neither the history ablation nor the Answer-3Seed control shows that relation heterogeneity causes the portfolio gain. What the experiments do establish is that independently trained checkpoints provide

complementary candidates and that execution controls escalation, abstention, and the optional structural budget. The validation-execution selector converts that complementarity into a test-independent deployment choice. Its unconstrained winner matches Answer-3Seed on Seen and improves the mean Unseen budget frontier, while the explicit relation constraint is detrimental on Seen.

7 Discussion

7.1 What Produces the Observed Improvement?

Adapter training matters because every trained portfolio beats a zero-shot model that already receives trajectory context, and sequential execution matters because composition beats every member. The stronger causal attribution is checkpoint diversity: Answer-3Seed exceeds the original relation-heterogeneous portfolio on Seen and ties it on Unseen. Relation-mined targets still change the patch-scope distributions, but the present controls do not establish them as a coverage mechanism. Our final method therefore lets validation execution select any three checkpoints and reports both the one-per-relation restriction and the seed-matched Answer portfolio as mandatory comparators. The selected CodeNet portfolio contains two Answer checkpoints and Progress, empirically rejecting the assumption that all three target relations must appear. Counterfactual histories paired with identical current code would further test when path information changes the appropriate repair rather than only its supervision source.

7.2 Coverage Versus Patch Size

LSGen demonstrates the value of same-problem correct programs, reaching higher unrestricted coverage with mean TED over four times ZPDPATCH's. Smaller edits do not prove pedagogical superiority. The budget gate instead gives edit size an operational meaning: ZPDPATCH repairs more inputs than LSGen at every tested budget through TED 80, whereas LSGen dominates when broad changes are allowed. This is a deployment coverage-intervention frontier, not a learner-outcome claim.

7.3 Implications for Educational Use

ZPDPATCH produces programs, not hints; deployment should expose diffs or derive a hint ladder rather than overwrite work. Its fallback prevents presenting observed-test regressions as help. A learner study measuring transfer, time, and help seeking is still required to establish educational benefit; the present contribution is a repair and feedback-authoring substrate.

8 Threats to Validity

Construct validity. Passing the available tests is a proxy for semantic correctness and may admit overfitting, a longstanding generate-and-validate repair risk [Qi et al. 2014; Smith et al. 2015]. We reduce this risk by requiring the recorded oracle to pass the same suite, but hidden tests could change absolute rates. AST TED measures structural change, not readability, conceptual difficulty, or educational value. "Reachable" means observed later in a trajectory; it does not imply that the generated transition lies within an individual learner's ZPD. We therefore avoid claims about learning outcomes.

Internal validity. Dataset construction, verdict ordering, normalization, runner limits, and candidate selection can affect results. We use one runner and shared outcome cache across methods, preserve full trajectories after filtering, and compare paired outputs on identical instances. LSGen required infrastructure adaptation; although its retrieval and generation logic is retained, our integration may differ from its original environment. Greedy decoding removes sampling variance, and

three full QLoRA runs quantify training-seed variation, but other hyperparameters and stochastic decoding remain untested.

Data leakage and dependence. Unseen problems are completely excluded from training and retrieval. This problem-level separation follows evidence that duplicated code across machine-learning splits can materially inflate code-model evaluation [Allamanis 2019]. Seen splits share problem identities by design, and users may occur across splits; the benchmark is not user-held-out. We address assignment-level dependence with problem-cluster bootstrap and an equal-problem-weight estimand. The Seen paired gain persists among 155 training-disjoint-user instances, but the smaller Unseen disjoint-user interval includes zero; a prospectively user-held-out split would more directly test personalization transfer.

External validity. The study covers Python solutions from Project CodeNet, one 7B code model, one GPU, and programming-contest-style tasks. Results may not transfer to multi-file projects, other languages, industrial defects, private autograders, or novice courses with different submission behavior. The Unseen set consists of lower-volume problems by construction and should not be treated as a random sample of all new assignments.

Conclusion validity. We define explicit metric families, correct planned RR comparisons, and repeat primary effect estimates with problem clusters. TED intervals remain conditional on joint successful repair and should not be generalized to failures. The current model family and testcase suite still delimit the measured effects; additional architectures and hidden tests would test whether the ordering transfers.

Ethical considerations. The study analyzes a publicly released code dataset and performs no intervention with learners. We do not attempt to re-identify users, and derived evaluation reports need only pseudonymous identifiers. Nevertheless, source-code histories can carry authorship signals; a public artifact should minimize identifiers and follow Project CodeNet’s distribution terms.

9 Conclusion

We presented ZPDPATCH, an execution-guided APR framework trained from authentic submission trajectories. Nested execution-evidence orders define auditable Progress and Strict targets, while terminal closure supplies Answer targets. Independent composition raises Seen RR from 50.1% for its strongest member to 59.5%, and a same-target Answer-3Seed portfolio further reaches 61.7%, showing that independent checkpoint diversity is the empirically supported coverage mechanism. The system outperforms trajectory-prompted zero-shot repair on Seen and problem-held-out Unseen data under problem-clustered inference. Same-problem retrieval has higher unrestricted coverage, while the certified edit-budget gate gives ZPDPATCH higher coverage through TED 80. Validation execution poses relation-constrained portfolio choice as exact submodular coverage maximization; the resulting test evaluation separates this method from post-hoc seed choice.

Data availability. An anonymized artifact will provide code, prompts, splits, outputs, and a SHA-256 manifest. Project CodeNet and CodeWorkout are reacquired from their official sources; deterministic regeneration and checksums are included.

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